

Kavya Mothukuri

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Education

University of Georgia, Masters in Computer Science (3.93/4.0)	Aug 2024 – May 2026
National Institute of Technology Raipur, Bachelors in IT (8.53/10.0)	Dec 2020 – May 2024

Experience

Graduate Research Assistant (Software and AI), University of Georgia, GA Jan 2025 – Present

- Collaborated with University of British Columbia and developed a biomechanical hand model in **Java**, increasing simulation accuracy by **30%** for stroke movement analysis.
- Designed and automated large-scale biomechanics simulation workflows on **AWS** (EC2, S3, CloudFormation), increasing computational throughput and research reliability.
- Developed, deployed, and maintain CodexLab's **full-stack MERN platform**, streamlining collaboration across projects and enhancing lab workflow efficiency.

Software Engineer Intern, OpenGenus – India Jun 2023 – Dec 2024

- Authored technical articles on algorithms and systems design for the OpenGenus Foundation, expanding its open-source learning resources.
- Designed and implemented a **LeetCode Visualizer** using **MERN, GraphQL, and Redis**, improving query performance by **50%**.
- Built interactive dashboards to visualize coding profiles, track LeetCode rating growth, and benchmark performance against peers.
- Developed analytics features to categorize solved problems by type and difficulty, enabling students to identify strengths and areas for improvement.
- Deployed the tool to production: **LeetCode Visualizer**.

Technical Skills

- **Programming Languages:** C, C++ , Python, Java, JavaScript, TypeScript
- **Web Development:** React.js, Redux, Angular, Node.js, Express.js, MongoDB, RESTful APIs, HTML5
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB
- **Coursework:** Data Structures, Algorithms, Operating Systems, Database Management, Software Engineering, Computer Science Fundamentals, OOPs
- **Cloud Platforms:** AWS (EC2, S3, RDS, Lambda, API Gateway, CloudFront, CloudFormation, IAM, VPC, Route 53, ELB, Auto Scaling, CloudWatch, DynamoDB, EKS, ECS), Azure, GCP
- **Cloud Architecture:** Infrastructure as Code (IaC), Serverless Architecture, Microservices Design, High Availability, Fault Tolerance, Scalability, Cost Optimization
- **Networking and Security:** VPC Design, Subnetting, Security Groups, NACLs, VPN, IAM, Cloud Security Best Practices, TCP, UDP, BGP, gRPC, TLS/SSL, HTTP/HTTPS, Compliance (SOC2, HIPAA)
- **DevOps and Methodologies:** CI/CD Pipelines, GitHub Actions, Docker, Kubernetes, Agile Methodologies

Projects

Serverless Note Taking App, Project Link.

- Built and deployed a serverless full stack-application using SST, AWS Lambda, DynamoDB, and React for scalable, cost-efficient infrastructure.
- Implemented authentication and authorization with AWS Cognito to ensure secure and personalized user access.
- Integrated API Gateway with DynamoDB to deliver low-latency operations, improving response times by 30%.
- Integrated Stripe for secure payment processing, enabling seamless subscription and transaction support for users to save their notes
- Optimized CI/CD pipeline for continuous testing and deployment, cutting release cycle time by 25%.
- Developed a responsive React frontend, improving engagement and accessibility across devices.

Tomato Leaf Disease Classification, Research Project

- Built a hybrid deep learning model combining LSTM and MobileNetV2 to classify tomato leaf diseases.
- Achieved 90% accuracy by capturing both spatial and temporal features for improved detection.
- Utilized Python, TensorFlow, Keras, OpenCV, and NumPy for model development and evaluation.
- Contributed to early plant disease diagnosis, supporting precision agriculture and crop yield improvement.